

Fenrix Strategy Validation Lab

Backtesting shows where a strategy worked. Fenrix searches for where it breaks.

- Product flow: Describe → Review → Lock → Backtest → Stress → Minimize → Export
- Strategy hash (immutable): c9d19b2e8d20667a...
- Source code SHA: 8b2140bdb9ff5333b6e457df79c20e43ac9f9738 · data mode of record: yfinance · universe: 30 assets

The problem

A green backtest is a claim, not a validation

- Backtests only tell you where a strategy already worked — they are silent about fragility.
- Students and retail quants ship strategies validated on one historical path.
- Overfitting, cost blindness and regime fragility hide until real capital finds them.
- There is no cheap, reproducible way to ask: under which conditions does this break?

Product workflow

Describe → Review → Lock → Backtest → Stress → Minimize → Export

- Describe: plain-English strategy → structured clause ledger (every clause reviewed).
- Review + Lock: mandatory approve step → immutable version + canonical SHA-256 hash.
- Backtest: the SAME locked hash runs the historical backtest on the run-of-record data (yfinance).
- Stress: the SAME hash enters a sealed synthetic failure search across mechanisms.
- Minimize: confirmed failures are shrunk to a minimal reproducible counterexample.
- Export: evidence package with manifest, hashes and claim ledger.

[← Strategy Lab](#) **Fenrix Submission MVP** source: — git: — hash: —

One story, eight verifiable stages: plain-English strategy → reviewed clause ledger → mandatory approval → immutable hash → declared data source → real multi-asset backtest → sealed synthetic stress → minimized failure + adjacent pass → signed evidence.

🔔 **New here? How to use this page**

1. **Pick a universe** — the list of tickers your strategy trades. For SAA/TAA class work, choose “**SAA / TAA building blocks**” (asset-class ETFs: stocks, bonds, gold, commodities...), or pick **Custom** and type your own comma-separated tickers.
2. **Optionally pick a strategy goal** — a plain-English starting point (60/40, risk parity, momentum). Note: these are goal descriptions; the engine always runs the flagship long/short momentum model. Its description appears below the controls.
3. Click “▶ **Run live pipeline**” to compile, approve, backtest and stress-test your setup end-to-end. Or click “✓ **Run verified judge demo**” to see a finished, canned example first.
4. **Read the 8 stages** as they light up, then **download the signed evidence JSON** at the end (stage 8).

Everything here is for research and education only — it is not investment advice.

Universe preset Flagship 29 (default)

Example strategy goal — not executed Flagship L/S momentum (default — the engine that actually runs)

Templates are plain-English starting points only. Every run executes the flagship long/short momentum engine, the SAA/TAA labels describe a goal, not a different allocator. Full SAA/TAA execution is on the roadmap.

Mode Tier 3 · deterministic synthetic (offline) Stress budget 8

▶ Run live pipeline ✓ Run verified judge demo

Flagship L/S momentum (default) · universe: Flagship 29 (default) — 30 tickers

Fenrix flagship long/short momentum-volatility strategy; rank the universe cross-sectionally by momentum and volatility, hold the top ranks long and bottom ranks short, rebalance on schedule with cost controls. This is what the pipeline executes.

1 · Strategy

2 · Clause review

3 · Approval

4 · Data source

5 · Historical results

6 · Sealed stress

7 · Failure replay

8 · Evidence export

Live log

idle...

Charts are hand-rolled inline SVG (zero external dependencies). The verified judge demo renders persisted, deterministic evidence with no live-server dependency. Synthetic stress worlds probe fragility, not real future risk; costs are labeled heuristics, not broker-calibrated.

Demo run — Real historical backtest (yfinance)

30 assets · 50 trades · all figures from deck_data.json

13.07%

cumulative return

0.18

Sharpe

-26.75%

max drawdown

14.65%

ann. volatility

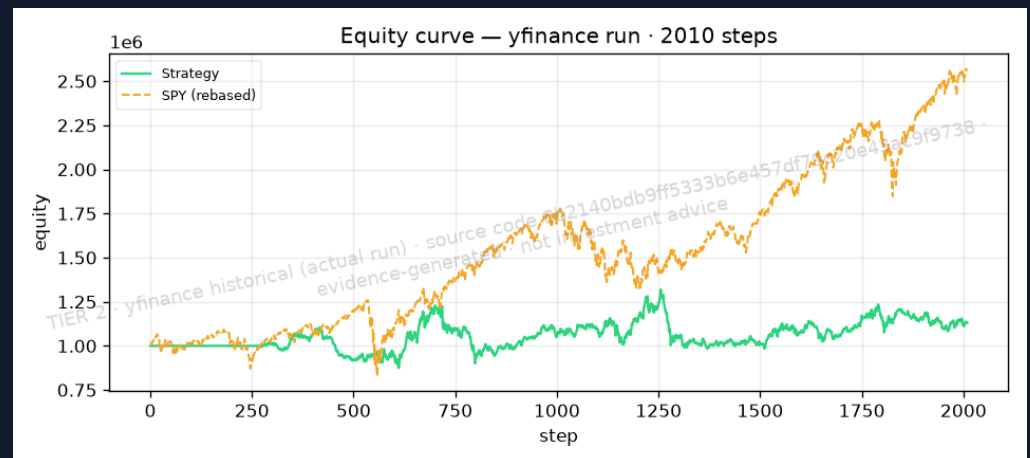
14.03%

cost % of capital

50

trades

- CAGR 1.55% · Sortino 0.22 · Calmar 0.06
- Benchmark CAGR 12.49% · information ratio -0.46
- Gross exposure avg 103.70% · net exposure avg 0.00%
- Costs are explicit bounded heuristics (commission/spread/slippage/borrow) — not broker calibrations.



What the normal backtest missed

Baseline vs confirmed stress failure

- Baseline (Real historical backtest (yfinance)): Sharpe 0.18, max drawdown -26.75%, cumulative return 13.07%.
- Sealed stress search: 8 confirmed failures out of 24 worlds (33.33% failure rate).
- Confirmed-failure mechanisms: borrow_cost_increase, correlation_breakdown, slippage_inflation, volatility_expansion.
- Stress-mechanism result matrix (per mechanism, base seed):
 - momentum_reversal: pass
 - volatility_expansion: FAIL (low_sharpe)
 - correlation_breakdown: FAIL (low_sharpe)
 - volatility_compression: pass
 - spread_inflation: pass
 - slippage_inflation: FAIL (low_sharpe)
 - borrow_cost_increase: FAIL (low_sharpe)
 - short_unavailability: pass
 - delayed_rebalance: pass
 - universe_churn: pass
 - missing_data_shock: pass

TIER 2

Minimized counterexample

Smallest reproducible world where the locked strategy still fails

- Baseline vs minimized failure — the breaking condition, shrunk to a minimal reproducible counterexample:
- • BASELINE (Real historical backtest (yfinance)): Sharpe 0.18, cum. return 13.07%.
- • MINIMIZED FAILING WORLD (volatility_expansion @ intensity 0.9285, seed 20264): still fails = True; violated predicates = low_sharpe.
- • Adjacent pass: no adjacent pass within radius — reported honestly, not fabricated.

TIER 2

Education & research value

A falsification lab, not a profit promise

- Students see WHY a strategy fails — mechanism, intensity, seed — not just an equity curve.
- Every claim is bound to a hash and an evidence file; grading and review become reproducible.
- Instructors can replay the minimized counterexample deterministically.
- Mechanism library searched this run: borrow_cost_increase, correlation_breakdown, delayed_rebalance, missing_data_shock, momentum_reversal, short_unavailability, slippage_inflation, spread_inflation, universe_churn, volatility_compression, volatility_expansion.

Architecture & moat

Immutable contracts + sealed synthetic worlds

- Clause ledger → canonical hash: the same immutable artifact flows through every stage.
- Sealed synthetic world generator: seeds + mechanisms are reproducible and tamper-evident.
- Evidence packages carry SHA-256 manifests binding data, backtest, campaign and replay.
- Failure minimizer + adjacent-pass search turn red flags into actionable counterexamples.
- Moat: the corpus of confirmed, minimized failures compounds with every run.

Honest boundary & roadmap

What this is NOT — and what comes next

- Synthetic stress worlds are generated, not historical; they probe fragility, not real future risk.
- Costs are explicit bounded heuristics (flat bps), not validated broker calibrations.
- Fenrix fundamentals (if used) are not point-in-time; lagged approximation only.
- No claim of live profitability, production execution fidelity, or universal realism.
- Failure search is not exhaustive; it reports confirmed failures only.
- Roadmap: point-in-time fundamentals, broker-calibrated costs, richer mechanism library, live paper-trading bridge.

Ask

Help us make strategy falsification the default

- Pilot with quant-finance classrooms and student funds this semester.
- Feedback on mechanism realism from practitioners.
- Compute credits for larger sealed stress campaigns.
- Reproduce everything: make submission-demo · make verify-submission · make pitch-deck.